SIMATS ENGINEERING

NATURAL LANGUAGE PROCESSING – ASSIGNMENT – 04

Part I

Semantic Analysis Through Structured Reasoning

Details	**Information**
Course	Natural Language Processing
Course Outcome	CO 4 – Analyse and apply semantic representation and interpretation techniques in NLP
Nature of Assignment	Individual
Duration	One Week
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Student Details

Field	**Details**
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Task 1 – Meaning Representation and Language Structure (8 Marks)

Scenario: A conversational assistant receives the sentence "Maya gave Ravi a book yesterday." The system must represent who did what, to whom, and when.

1.1 Identify Meaning Components

Component	**Analysis**
Entities	Maya – giver/Agent; Ravi – recipient/Goal; a book – transferred object/Theme.
Event	A giving/transfer event.
Participants	Maya, Ravi, and the book participate in the event.
Semantic roles	Maya = Agent; Ravi = Recipient/Goal; book = Theme.
Temporal information	Yesterday = time of the giving event.

Linguistic form	Subject + Verb + Indirect Object + Direct Object + Temporal Adverbial.
Meaning	There exists a past giving event involving Maya, Ravi, and a book, occurring yesterday.

1.2 Compare Meaning Representations

The same meaning can be represented at different levels. A conceptual/event representation focuses on the event and its participants, while a frame representation organizes the information into named slots.

A simple event structure:

GIVE(e1) ; AGENT(e1, Maya) ; RECIPIENT(e1, Ravi) ; THEME(e1, Book) ; TIME(e1, Yesterday)

Structured frame:

Slot	**Value**
Frame	GIVING
Event ID	e1
Agent	Maya
Recipient	Ravi
Theme	Book
Time	Yesterday

The event structure makes relations explicit and is convenient for reasoning. The frame makes the role slots easy to access in a software system. Both leave some details implicit, such as the exact identity of the book and the precise clock time.

1.3 Meaning Structure and Ambiguity

Word order helps a system identify grammatical roles, but semantic roles are not determined only by word position.

Original:

Maya gave Ravi a book yesterday.

Alternative word order:

Yesterday, Maya gave a book to Ravi.

Both have approximately the same semantic representation:

GIVE(e1)

Agent(e1, Maya)

Recipient(e1, Ravi)

Theme(e1, Book1)

Time(e1, Yesterday)

The words are arranged differently, but the underlying semantic relationships remain equivalent.

1.4 Design Defence

A structured event/frame representation is suitable for a computer because entities, roles and relations are stored as explicit fields rather than hidden in a paraphrase. The representation supports querying, reasoning, information extraction and retrieval. For example, the system can directly answer who received the book or when the event happened. It can also be extended with confidence, entity identifiers and additional relations.

Task 2 – First Order Predicate Calculus (8 Marks)

2.1 Define the Logical Vocabulary

Constants

- Rao – Dr. Rao
- NLP – a particular course

Variables

- x, y – students, instructors or courses depending on context

Symbol	Type / Meaning	Argument Order
Student(x)	x is a student	Unary
Course(c)	c is a course	Unary
Instructor(i)	i is an instructor	Unary
Enrolled(x,c)	student x is enrolled in course c	Student, Course
Teaches(i,c)	instructor i teaches course c	Instructor, Course
Passed(x,c)	student x passed course c	Student, Course
NLP	Constant denoting the NLP course	Constant
DrRao	Constant denoting Dr. Rao	Constant

2.2 Translate Natural Language into Logic

(i) Every student is enrolled in at least one course.

$\forall x\ [Student(x) \rightarrow \exists y\ (Course(y) \wedge Enrolled(x,y))]$

Meaning: For every student, there exists at least one course in which that student is enrolled.

(ii) Some student passed every course taught by Dr. Rao.

$\exists x\ [Student(x) \wedge$

$\quad \forall y\ ((Course(y) \wedge Teaches(Rao,y)) \rightarrow Passed(x,y))]$

Meaning: There is at least one student who passed every course taught by Dr. Rao.

(iii) No instructor teaches every course.

$\forall x\ [Instructor(x) \rightarrow$

$\quad \neg\forall y\ (Course(y) \rightarrow Teaches(x,y))]$

An equivalent form is:

$\forall x\ [Instructor(x) \rightarrow$

$\quad \exists y\ (Course(y) \wedge \neg Teaches(x,y))]$

The second form directly says that every instructor has at least one course that they do not teach.

2.3 Interpret and Check the Formulae

Consider:

$\exists x\ [Student(x) \wedge$

$\quad \forall y\ ((Course(y) \wedge Teaches(Rao,y)) \rightarrow Passed(x,y))]$

Scope

The existential quantifier $\exists x$ has scope over the complete formula inside it.

The universal quantifier $\forall y$ has scope over:

$(Course(y) \wedge Teaches(Rao,y)) \rightarrow Passed(x,y)$

In simple language:

There is one particular student who passed every course taught by Rao.

Quantifier Order

Compare:

$\exists x \forall y$ Passed(x,y)

with:

$\forall y \exists x$ Passed(x,y)

The first means:

There is one student who passed every course.

The second means:

For every course, there may be a different student who passed it.

Counterexample Model

Suppose there are two students:

Alice

Bob

and two courses:

NLP

DBMS

Suppose:

Alice passed NLP

Bob passed DBMS

Then:

$\forall y \exists x$ Passed(x,y)

can be true because every course has some student who passed it.

But:

$\exists x \forall y$ Passed(x,y)

is false because neither Alice nor Bob passed both courses.

2.4 Logical Consistency

Statements:

Every student passed NLP.

There is a student who did not pass NLP.

The formulas are:

$\forall x$ [Student(x) $\rightarrow$ Passed(x,NLP)]

and

$\exists x$ [Student(x) $\wedge \neg$Passed(x,NLP)]

They **cannot both be true**.

The second statement provides a student who did not pass NLP, contradicting the first statement that every student passed NLP.

Task 3 – Linguistically Relevant Concepts and Syntax-Driven Semantic Analysis (9 Marks)

Sentence: "The experienced researcher saw the student with the telescope."

3.1 Identify Linguistically Relevant Concepts

Concept	**Representation / Role**
Researcher	Entity; experienced modifier applies to researcher.
Student	Entity; object of saw.
Telescope	Entity used as an instrument or possible accompaniment.
Saw	Perception event; researcher is Experiencer and student is perceived entity.
With the telescope	PP with ambiguous attachment.
Semantic relations	SEE(e, researcher, student); INSTRUMENT(e,telescope) in the instrument reading.

Important concepts include:

Researcher – subject/experiencer-like participant

Student – object/theme of seeing

Telescope – possible instrument

Experienced – adjective modifying researcher

Saw – perception event

These concepts are necessary because the same words can produce different meanings depending on their semantic relationships.

3.2 Syntax-to-Semantics Mapping

Attachment A – VP attachment: “The experienced researcher saw [the student] [with the telescope].” The PP modifies the seeing event. Semantic reading: the researcher used the telescope while seeing the student.

SEE(e1, Researcher, Student) ∧ INSTRUMENT(e1, Telescope)

Attachment B – NP attachment: “The experienced researcher saw [the student with the telescope].” The PP modifies the noun phrase student. Semantic reading: the student is associated with/has the telescope.

SEE(e1, Researcher, Student) ∧ WITH(Student, Telescope)

Syntax-driven procedure: build the parse tree → identify the head of each phrase → compose the semantic representation of the head with its complements/modifiers → produce candidate sentence meanings → compare candidates using context.

3.3 Semantic Composition

The interpretation can be built in stages.

Step 1 – Noun meanings

researcher → RESEARCHER(x)

student → STUDENT(y)

telescope → TELESCOPE(z)

Step 2 – Adjective

experienced researcher

becomes:

RESEARCHER(x) ∧ EXPERIENCED(x)

Step 3 – Verb

saw

introduces a seeing event:

SEE(e,x,y)

Step 4 – Object

the student

fills the Theme/Object role:

Theme(e,y)

Step 5 – PP

If attached to VP:

Instrument(e,z)

If attached to NP:

HAS(y,z)

Therefore, the interpretations diverge at the **attachment of the PP**.

3.4 Failure Analysis

Error	**Consequence**	**Validation Check**
Wrong PP attachment	System may claim the student owns/has the telescope when it was actually used as an instrument.	Check event-role compatibility and use context such as "looked through the telescope."
Assigning roles from word order alone	A PP or passive construction may be assigned the wrong semantic role.	Use dependency/constituency structure plus a role-compatibility check against the verb's frame.

Task 4 – Lexemes, Senses and Word Sense Disambiguation (9 Marks)

4.1 Lexemes and Senses

A lexeme is an abstract vocabulary item associated with a family of word forms and senses. A word form is the actual surface form appearing in text. A sense is a particular meaning of a lexeme in context.

Context	**Word form**	**Lexeme / grammatical use**	**Relevant sense**
A: Nila deposited cash	bank	BANK noun	Financial institution

at the bank.			
B: They sat on the bank of the river.	bank	BANK noun	River edge / riverside
C: The pilot banked the aircraft to the left.	banked	BANK verb	Tilt/turn an aircraft laterally

The noun uses in A and B share the word form “bank” but have different senses. C is a past-tense verb form of the verbal lexeme BANK.

4.2 Design a Context-Based WSD Method

Input

A sentence containing the ambiguous word bank.

Output

The most probable sense of bank.

Candidate Senses

S1 = Financial institution

S2 = River edge

S3 = Aircraft movement/tilting

Context Features

Use:

Words occurring near bank

Part of speech

Verb-object relationships

Semantic category of nearby nouns

Prepositions

Domain indicators

Decision Procedure

Step 1: Identify the target word.

Step 2: Determine its part of speech.

Step 3: Generate candidate senses.

Step 4: Extract nearby context words.

Step 5: Compare context with the semantic features of each sense.

Step 6: Select the sense with the highest compatibility score.

Example:

cash + deposited + bank

strongly supports the financial sense.

4.3 Test the Method

Sentence	Evidence	Selected Sense
Nila deposited cash at the bank.	deposited, cash, at; financial vocabulary	FINANCIAL
They sat on the bank of the river.	sat, river, on the bank	RIVER-EDGE
The pilot banked the aircraft to the left.	pilot, aircraft, banked, left	AIRCRAFT-TURN
The company asked for a bank loan.	company, loan, financial context	FINANCIAL

Possible failure: “The bank was crowded” is ambiguous without context because both a financial institution and a river bank can be crowded. A larger context window or a knowledge-based clue is needed.

4.4 Evaluation

Create a labelled test set containing sentences for each sense, for example 30 sentences with 10 examples per sense. Compare predicted labels with gold labels.

Accuracy = correctly classified examples / total examples.

If 27 of 30 examples are correct, accuracy = 27/30 = 90%. A balanced test set can hide real-world weaknesses because actual corpora may have highly unequal sense frequencies, domain shifts, rare

senses and new contexts. Therefore, per-sense recall, confusion matrices and domain-specific evaluation should also be reported.

Task 5 – Information Retrieval and Semantic Matching (8 Marks)

5.1 Query and Document Representation

Query: "methods for finding the meaning of ambiguous words."

Element	Semantic interpretation
Central concept 1	Word meaning / lexical semantics
Central concept 2	Ambiguity / multiple senses
Central concept 3	Methods for disambiguation
Likely intent	Find techniques or algorithms for determining the intended sense of an ambiguous word.
Query representation	Concept vector / embedding + extracted concepts such as WSD, lexical ambiguity, word senses.
Document representation	Title, abstract, content concepts, embeddings, keywords and domain metadata.

5.2 Lexical versus Semantic Retrieval

Method	Strength	Weakness
Keyword matching	Simple, transparent and efficient when query and document share terms.	May miss documents using synonyms or different terminology.
Semantic retrieval	Matches concepts and related meanings even when exact words differ.	May retrieve related but non-answering documents if semantic similarity is too broad.

Example missed by keyword retrieval: a document titled "Determining the Intended Sense of Polysemous Terms Using Context" may discuss WSD without using the exact phrase "meaning of ambiguous words." Semantic retrieval can connect polysemy, context and sense selection.

5.3 Ranking Design

Use a weighted ranking score:

$$\text{Score}(d,q) = 0.35 \times \text{TermRelevance} + 0.35 \times \text{ConceptSimilarity} + 0.20 \times \text{SenseMatch} + 0.10 \times \text{FieldWeight}.$$

Signal	Effect
Term relevance	Rewards important query terms and avoids overvaluing common words.
Concept similarity	Rewards semantic closeness between query and document.
Sense match	Rewards documents explicitly addressing ambiguity and word senses.
Field weighting	Gives additional importance to title/abstract matches.

5.4 Evaluation and Error Analysis

Precision = relevant retrieved documents / all retrieved documents. Recall = relevant retrieved documents / all relevant documents.

Document	Retrieved?	Relevant?	Classification
D1 – WSD methods	Yes	Yes	TP
D2 – lexical semantics	Yes	Yes	TP
D3 – general NLP introduction	Yes	No	FP
D4 – information retrieval	No	Yes	FN
D5 – machine translation	No	No	TN

If 2 relevant documents are retrieved out of 3 retrieved documents, Precision = 2/3 = 66.7%. If there are 3 relevant documents in the collection and 2 are retrieved, Recall = 2/3 = 66.7%.

A semantic retrieval error occurs when a document discusses “word embeddings” and is highly similar to the query but does not actually explain methods for disambiguating word senses. A sense-aware relevance check can reduce this error.

Task 6 – Integrated Semantic Analysis Challenge (8 Marks)

Message: “I saw her duck near the bank, and it was closed.”

This sentence contains several ambiguities.

Stage 1 – Interpret the Problem

Ambiguity 1: duck

It can be:

A noun: an animal

A verb: lower one's head/body

Ambiguity 2: bank

It can mean:

Financial institution

River bank

Ambiguity 3: it

The pronoun may refer to:

The bank

Another previously mentioned entity

Ambiguity 4: closed

Most naturally describes a business/place such as a bank, but context is needed.

Stage 2 – Build Candidate Interpretations

Interpretation 1 – Duck as an Animal

Meaning:

I saw her duck near the river bank, and the river bank was closed.

Semantic representation:

SEE(e1)

Agent(e1, Speaker)

Theme(e1, Duck1)

Location(e1, NearBank1)

DUCK(Duck1)

RIVER_BANK(Bank1)

CLOSED(Bank1)

Possible interpretation:

Speaker saw a duck near a river bank.

The river bank was closed.

However, the pronoun resolution depends heavily on context.

Interpretation 2 – Duck as a Verb

Meaning:

I saw her lower herself near a bank, and the bank was closed.

Representation:

SEE(e1)

Agent(e1, Speaker)

Experiencer(e1, Her)

DUCK(e2)

Agent(e2, Her)

Location(e2, NearBank1)

BANK(Bank1)

CLOSED(Bank1)

Here duck functions as a verb, describing an action performed by her.

If bank refers to a financial institution, the interpretation could be:

I saw her duck near the bank, and the bank was closed.

This is quite plausible in a real-world context.

Stage 3 – Syntax-Driven Analysis and WSD

The system should first examine possible syntactic structures.

Candidate A

her duck

is treated as a noun phrase:

her + duck(noun)

Meaning:

the duck belonging to/associated with her.

Candidate B

her duck

is interpreted with duck as a verb:

her + duck(verb)

Meaning:

she lowers herself.

The system then uses context.

For bank

If nearby context contains:

money

account

loan

cash

ATM

select:

BANK = financial institution

If it contains:

river

water

shore

fishing

select:

BANK = river edge

Most Useful Additional Context

The best additional information would be:

Previous sentence

Following sentence

Location

Whether the conversation concerns banking

Whether a duck/animal has already been mentioned

Whether a bank is physically nearby

Stage 4 – Information Retrieval Trace

Selected interpretation: bank = financial institution; duck = verb or bird depending on the syntactic parse; "closed" is a strong clue that the bank institution is the relevant entity.

1. Extract concepts: bank closure, financial institution, opening hours, branch access.
2. Normalize terms: closed → closure/closed branch; bank → financial institution.
3. Construct semantic query: {financial institution, bank branch, closed, opening hours, access} plus context embeddings.
4. Retrieve candidate articles using lexical and semantic similarity.
5. Apply sense filter: reject documents about river banks unless other evidence supports that sense.
6. Rank remaining documents using term relevance, concept similarity and sense match.

Example ranking:

Article	**Bank Sense**	**Topic Match**	**Rank Decision**
Branch closed – opening hours	Financial	High	1
Why river banks flood	River	Low	Reject
Bank holiday schedule	Financial	High	2

Stage 5 – Challenge and Refine

Misleading Context

Suppose the previous message says:

"We went for a walk beside the river."

The system may incorrectly choose:

bank = river bank

because river is strongly associated with that sense.

However, the phrase:

“it was closed”

may indicate a bank branch, especially if the conversation is about banking services.

Refinement

Add cross-sentence context and semantic-role compatibility.

The system should combine:

Local context

+

Previous sentences

+

Syntactic structure

+

World knowledge

+

Sense compatibility

instead of relying on a single nearby word.

Conclusion

The assignment demonstrates that semantic analysis is more than replacing words with dictionary definitions. A robust NLP system must connect linguistic structure with explicit meaning representations, logical reasoning, word-sense decisions and retrieval objectives. Event structures and frames provide machine-readable semantic roles; FOPC supports quantifier-based reasoning; syntax-driven composition handles attachment ambiguity; WSD separates lexemes from context-specific senses; and semantic retrieval connects user intent to documents even when wording differs.

For the integrated helpdesk scenario, the strongest design is a pipeline that combines syntactic parsing, compositional semantics, WSD, coreference resolution and sense-aware retrieval. The system should preserve alternative interpretations when evidence is insufficient rather than silently selecting an unreliable meaning.